

Wedge

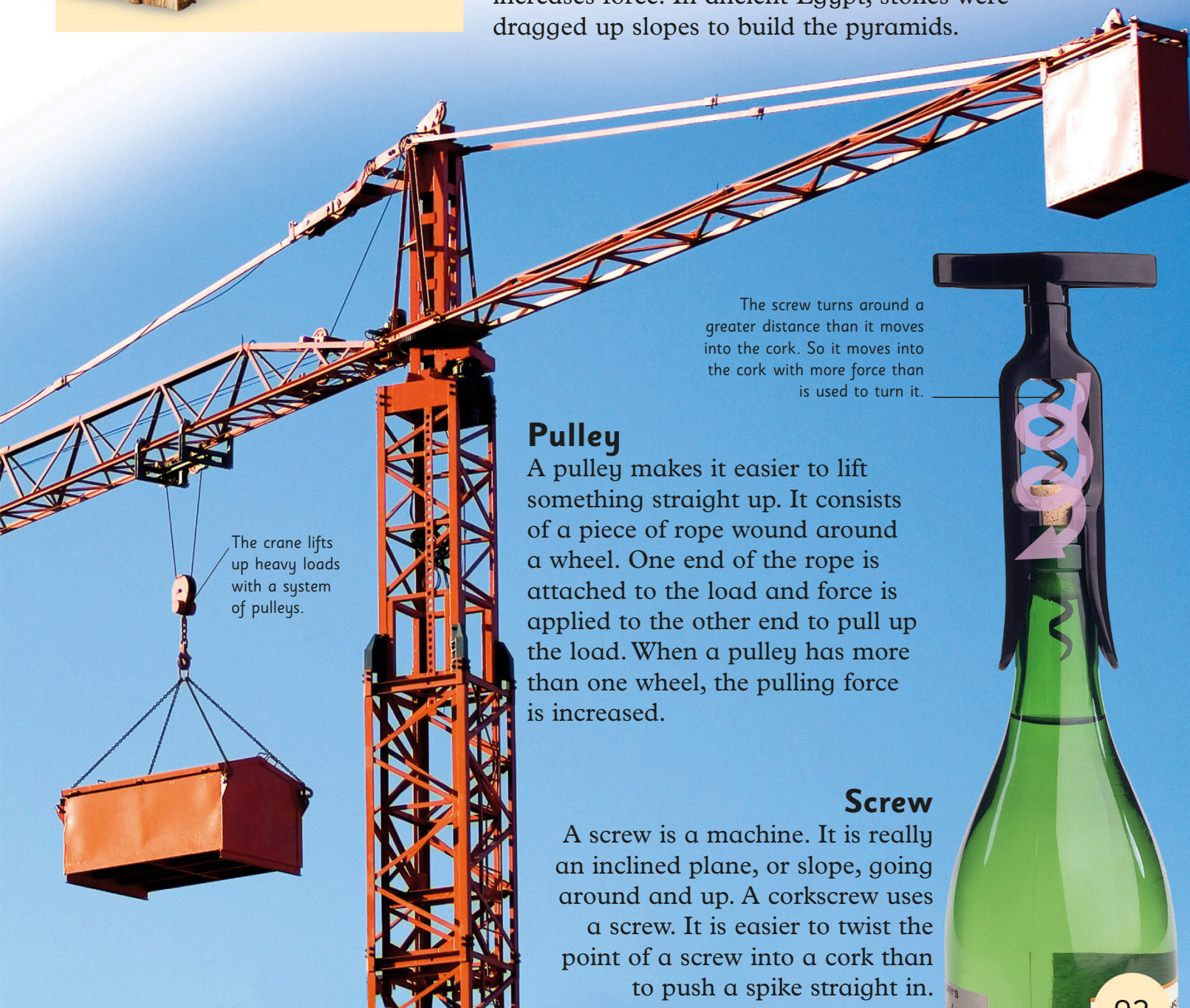
An ax blade is an efficient but simple machine that increases force. When it hits the wood, the wedge forces the wood to split apart between its fibers.

It takes just one man to pull a stone up the slope, but four men are needed to lift a stone straight up.



Inclined plane

It is easier to push or pull something up a slope than lift it straight up. A slope, or inclined plane, therefore increases force. In ancient Egypt, stones were dragged up slopes to build the pyramids.



The crane lifts up heavy loads with a system of pulleys.

The screw turns around a greater distance than it moves into the cork. So it moves into the cork with more force than is used to turn it.

Pulley

A pulley makes it easier to lift something straight up. It consists of a piece of rope wound around a wheel. One end of the rope is attached to the load and force is applied to the other end to pull up the load. When a pulley has more than one wheel, the pulling force is increased.

Screw

A screw is a machine. It is really an inclined plane, or slope, going around and up. A corkscrew uses a screw. It is easier to twist the point of a screw into a cork than to push a spike straight in.

